

Biology content

Content overview

Students are expected to demonstrate and apply the knowledge, understanding and skills described in the content. They are also expected to analyse, interpret and evaluate a range of scientific information, ideas and evidence using their knowledge, understanding and skills.

To demonstrate their knowledge, students should be able to undertake a range of activities, including the ability to recall, describe and define, as appropriate.

To demonstrate their understanding, students should be able to explain ideas and use their knowledge to apply, analyse, interpret and evaluate, as appropriate.

Students should develop their ability to apply mathematical skills to biology throughout the course. These mathematical skills include the ability to change the subject of an equation, substitute numerical values and solve algebraic equations using decimal and standard form, ratios, fractions and percentages. Further details of the skills that should be developed are given in *Appendix 6: Mathematical skills and exemplifications*. Students should also be familiar with *Système Internationale d'Unités* (SI) units and their prefixes, be able to estimate physical quantities and know the limits of physical measurements.

Practical work is central to any study of biology. For this reason, the specification includes 18 core practical activities that form a thread linking theoretical knowledge and understanding to practical scenarios. In following this thread, students will build on practical skills learned at GCSE (or equivalent), becoming confident practical biologists, handling apparatus competently and safely. Using a variety of apparatus and techniques, they should be able to design and carry out both the core practical activities and their own investigations, collecting data which can be analysed and used to draw valid conclusions.

Questions in examination papers will aim to assess the knowledge and understanding that students gain while carrying out practical activities, within the context of the 18 core practical activities, as well as in novel practical scenarios. Success in questions that indirectly assess practical skills will come more naturally to those candidates who have a solid foundation of laboratory practice and who, having carried out practical skills, have a thorough understanding of practical techniques. Where possible, therefore, teachers should consider adding additional experiments to the core practical activities.

Candidates will be assessed on their practical skills in papers 3 and 6. These papers will include testing the skills of candidates in planning practical work – both in familiar and unfamiliar applications – including risk management and the selection of apparatus, with reasons.

When data handling, candidates will be expected to use significant figures appropriately, to process data and to plot graphs.

Students should be encouraged to use information technology throughout the course.

Units

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Unit 1: Molecules, Diet, Transport and Health

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit includes a consideration of molecules that are important in biology – including water, carbohydrates, lipids, proteins and nucleic acids, providing a basis for many areas of biology. This leads to the relevance of diet to health and the cardiovascular system in particular. The unit includes cell membrane transport processes, such as diffusion and active transport, proteins, enzymes and protein synthesis. This unit also includes an understanding of the genetic code and how mutations can result in disorders. Students will also consider techniques for genetic screening and the associated ethical and social issues.

Practical skills

In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic. There are also opportunities to carry out additional practical work, such as investigating the structure of a mammalian heart.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit, including tabulation and graphical treatment of data, concentrations and using appropriate units for physical quantities, calculation of percentage changes, and finding the initial rate of enzyme-catalysed reactions. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Assessment information

- First assessment: January 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 80 marks.
 - Candidates must answer all questions.
 - The paper may include multiple-choice, short-open, open-response, calculations and extended-writing questions.
 - The paper will include a minimum of 8 marks that target mathematics at Level 2 or above.
 - Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
 - Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.
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